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Titolo da decidere

Sotto titolo

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Effective Sample Size (ESS) (1)

MCMC algorithms generate correlated samples $\theta_1, \theta_2, \dots, \theta_M \sim p(\theta)$. Because these samples are dependent:

$$1000 \text{ MCMC samples} \neq 1000 \text{ independent samples}$$

ESS estimates how many independent samples contain the same information as the M autocorrelated samples:

$$\text{ESS} = \frac{M}{1 + 2\sum_{s=1}^{M-1}\left(1 - \frac{s}{M}\right)\rho_s(f(\theta))}$$

- M: total number of generated samples
- ρ_s: autocorrelation of the chain at lag s
- $\left(1 - \frac{s}{M}\right)$: finite-sample weight factor

Effective Sample Size (ESS) (2)

ESS depends on the specific expectation $f(\theta)$ being estimated:

To evaluate samplers on high-dimensional targets (e.g., 250D Gaussian), the authors of the NUTS paper compute two separate ESS values for each dimension j:

- **First Moment (Mean):** $f(\theta) = \theta_j \implies \text{ESS}_j^{(\text{mean})}$
  - Measures efficiency in estimating the central location.
- **Second Central Moment (Variance):** $f(\theta) = (\theta_j - \mu_{\text{true}})^2 \implies \text{ESS}_j^{(\text{variance})}$
  - Measures efficiency in capturing uncertainty and scale.
  - Requires a long baseline run (e.g., 50,000 samples) to obtain μ_{true}.

Modern ESS diagnostics

Modern tools (PyMC / ArviZ) default to rank-normalized ESS (Vehtari et al., 2021):

- **Bulk ESS** (`"bulk"`): Rank-based measure of sampling efficiency in the center.
- **Tail ESS** (`"tail"`): Efficiency for extreme quantiles (5% and 95%).

Our Methodology: Despite modern defaults, in this project we use $\textbf{ESS}^{(\textbf{mean})}$ and $\textbf{ESS}^{(\textbf{variance})}$ to remain strictly coherent with the benchmark in the original NUTS paper (Hoffman & Gelman, 2014).

Aggregating ESS

Since ESS is computed per scalar component, we compute a single ESS separately for each dimension. We then aggregate:

1. **Joint Multi-Chain Calculation:** All parallel chains (e.g., 4 chains) are fed into the estimator simultaneously to penalize poor mixing, yielding a value of both $\mathrm{ESS}_j^{(\mathrm{mean})}$ and $\mathrm{ESS}_j^{(\mathrm{variance})}$ for each dimension
2. **Conservative Dimension Reduction:**
   - *Per dim. j:* Take the minimum of mean and variance ESS:

     $$\mathrm{ESS}_j = \min\left(\mathrm{ESS}_j^{(\mathrm{mean})}, \mathrm{ESS}_j^{(\mathrm{variance})}\right)$$

   - *Across all dimensions:* Take the worst-case dimension:

     $$\mathrm{ESS}_{\mathrm{final}} = \min_{j \in \{1,\ldots,250\}} \mathrm{ESS}_j$$

Result: A single, conservative metric reflecting the worst-explored feature of the distribution.

250-Dimensional Multivariate Normal

The Dataset & Target Distribution

1. This synthetic experiment directly samples from a high-dimensional target distribution $p(\theta)$
2. $\theta \in \mathbb{R}^{250}$ is a 250-dimensional parameter vector with a zero-mean multivariate Gaussian target distribution:

$$p(\theta) \propto \exp\left(-\frac{1}{2}\theta^T A\theta\right)$$

Our goal: sample efficiently from this correlated 250D target space.

Precision Matrix

The target precision matrix $A = \Sigma^{-1}$ dictates the geometry of the target space:

- A is generated randomly from a **Wishart distribution** $\mathcal{W}_D(I, \mathrm{df} = 250)$ with identity scale matrix I and 250 degrees of freedom.
- **Role of Wishart:** Guaranteed to yield a symmetric, positive-definite matrix that creates strong correlations and complex, non-isotropic geometry.
- To ensure fair comparison across all runs and samplers, the exact same matrix A is fixed across all experiments.

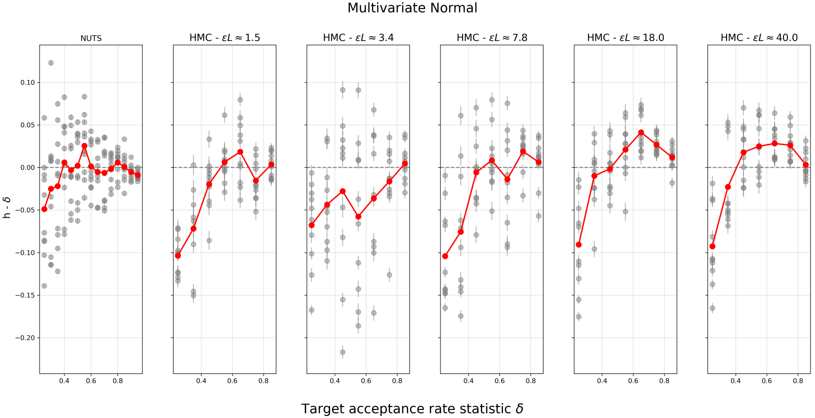
High Correl. + High Dim. = Ideal Benchmark for HMC vs. NUTS

Baseline Parameter Estimation

To compute $\text{ESS}^{(\text{variance})}$, the true parameter expectation $f(\theta) = (\theta - \mu_{\text{true}})^2$ requires a reference mean μ_{true}:

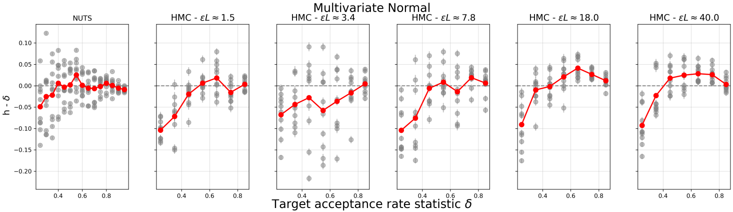
- We executed an initial baseline run using NUTS for **50,000 iterations** (with `target_accept` $= 0.5$).
- The long run ensures deep exploration and provides a highly precise estimate for $\mu_{\text{true}} \approx E[\theta] \approx \mathbf{0}$.

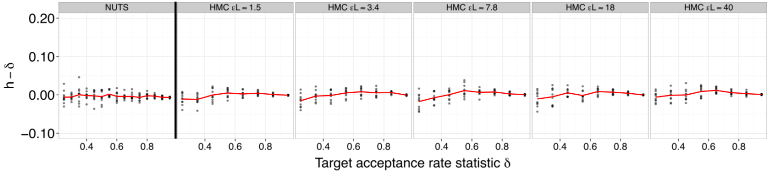
Target Acceptance (δ)



Each gray point is obtained by averaging across all 4 chains. The computation has been repeated 10 times for each value of δ. In red the average across all repetitions.

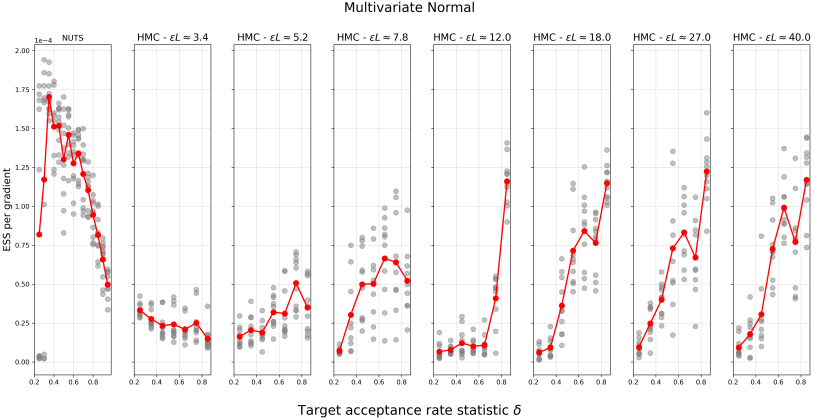
Target Delta (δ) comparison





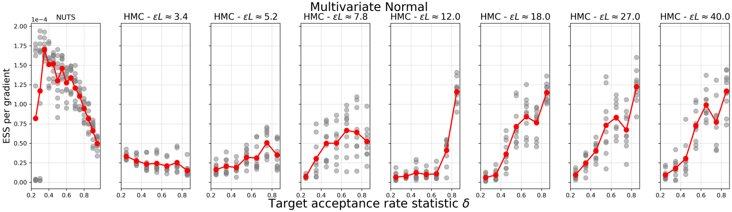
Above: our results for the 250D Gaussian. Below: the original paper results.

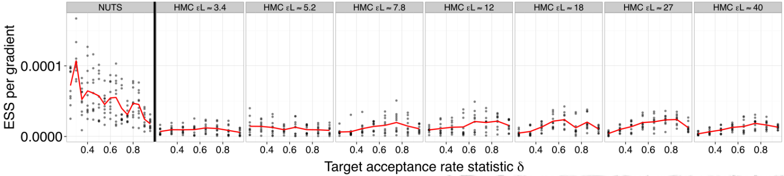
ESS per gradient



Each gray point is obtained by averaging across all 4 chains. The computation has been repeated 10 times for each value of δ. In red the average across all repetitions.

ESS per gradient comparison





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